

Preface

This volume contains the papers presented at the 16th workshop on Job Scheduling Strategies for Parallel Processing (JSSPP) that was held in Shanghai, China, on May 25, 2012, in conjunction with the IEEE International Parallel Processing Symposium 2012.

This year 24 papers were submitted to the workshop. All submitted papers went through a complete review process, with the full version being read and evaluated by an average of four reviewers. We would like to especially thank the Program Committee members and additional referees for their willingness to participate in this effort and their detailed, constructive reviews.

This workshop saw the crystallization of a trend in the parallel scheduling landscape. In addition to papers discussing traditional JSSPP topics like parallel batch scheduling, workload analysis and modeling, and resource management system software studies, Web scheduling emerged as a new topic this year. This volume includes a paper summarizing Walfredo Cirne’s keynote on Web-scale scheduling at Google. This paper provides a high-level overview of Web-scale scheduling workloads, as well as an approach that Google uses to meet service availability SLAs. The paper represents an early example of a broad class of scheduling problems impacting so-called “web scale” service providers.

In addition to this topic, scheduling issues were discussed in a broader context in more established areas, from hardware scheduling, to scheduling within budget constraints, scheduling for performance, and analysis of scheduling tasks within resource management software. Adopting a broader basis for scheduling discussions was an explicit goal this year for the workshop, and will be continued in future workshops.

A major goal of the JSSPP workshop is to explore new applications of scheduling approaches to novel scenarios. Many of papers presented this year attacked new schedule fitness metrics. In presenting their DEMB system, Lee et al. developed approaches to partition query parameter spaces in order to improve query locality in a distributed pool of servers. Wu et al. presented the Critical Path-base Priority Scheduling (CPPS) algorithm that improves end-to-end performance for DAG scheduling. Tian et al. demonstrated a series of algorithms that improve makespan of jobs on chip multiprocessors (CMPs).

In a more traditional parallel job scheduling vein, Klusáček et al. offered a detailed analysis of the fairness impact of a variety of popular scheduling algorithms. On the basis of this assessment, they have built an extension to conservative backfilling, using Tabu search, that performs explicit optimization for fairness as a part of the backfilling process. This approach represents an interesting addition of optimization into an often rigid phase of the scheduling process. Krakov et al. performed a high-resolution analysis of workload data, including the use of not only job workloads, but also the actual schedule on the system

as well. This comparison shows several discrepancies between these schedules and the results of simulation with commonly used scheduling algorithms. This finding suggests that simulation results may not be quite as comparable to real system performance as previously assumed. Zhang et al. analyzed 12 months of workload data from the Kraken petascale system. Niu et al. tackled the problem of inaccurate runtime estimates through the application of checkpointing to aggressive backfilling. This approach results in greatly improved system performance with low checkpointing overhead. Zakay et al. comprehensively discuss several alternative approaches to determine user session boundaries in workload data. This problem is particularly important to supercomputing centers because user sessions signal modal shifts in user expectations for scheduling.

Two papers analyzed the performance of production-grade resources that managers often used on extreme scale systems. Georgiou et al. analyzed the performance of the SLURM resource manager using a variant of NERSC's ESP toolkit for load testing. This work highlighted the difficulties in conducting benchmarking at large scales, and presented several techniques to project performance using combinations of full-scale testing and emulation. Bresford et al. discussed improvements to the LoadLeveller resource manager needed to accommodate the Blue Waters system. In order to scale to a system of this size, resource allocation was reworked to be a hybrid distributed process where node local resources are allocated at the node level. This approach trades scalability for increased communication. This paper also provides a detailed case study of improving the scalability of a production resource manager, detailing several performance bottlenecks and respective mitigation strategies.

Over the last few years, it has become clear there is a profound shift occurring in the scheduling landscape. While supercomputer workloads remain a substantial consumer of scheduling technology, and are driving the field in terms of scheduling heterogeneous resources and optimizing end-to-end behavior, it has become clear that these use cases are not the only ones in need of sophisticated scheduling approaches. Cloud schedulers that support interactive Web applications and services now manage larger quantities of resources in aggregate than traditional supercomputing centers and smaller HPC systems. As scheduling needs have expanded to include both interactive and hybrid workloads, new challenges have emerged. Therefore, we strongly believe that research in the field of this workshop will remain interesting and challenging for years to come.

The proceedings of previous workshops are available from Springer as LNCS volumes 949, 1162, 1291, 1459, 1659, 1911, 2221, 2537, 2862, 3277, 3834, 4376, 4942, 5798, and 6253. Since 1995 these volumes have also been available online.

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